

MH2500 Probability and Introduction to Statistics

Slides for Week 5-2

Topics: Continuous random variables

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We know this approximation

The Poisson distribution can be derived as the limit of a binomial distribution with $np = \lambda$, as $n \rightarrow \infty$.

Approximation is usually good when $n \geq 50$ and $np \leq 5$.

Example

An airline knows that overall 3% of passengers do not turn up for flights. Consequently, their policy is to sell 150 tickets for an aircraft with only 147 seats.

Find the probability that every person who shows up for the flight will find a seat available.

Solution: Let X be the number of people who do not show up, then

$$X \sim \text{Bin}(150, 0.03).$$

We want to find $P\{X \geq 3\}$.

- By $X \sim \text{Bin}(150, 0.03)$,

$$\begin{aligned} P\{X \geq 3\} &= 1 - P\{X \leq 2\} \\ &= 1 - \sum_{i=0}^2 \binom{150}{i} (0.03)^i (0.97)^{150-i} = 0.8307. \end{aligned}$$

- Approximate by Poisson: $Y \sim \text{Poisson}(4.5)$, where we note $150 \times 0.03 = 4.5 < 5$.

$$\begin{aligned} P\{X \geq 3\} &= 1 - P\{X \leq 2\} \approx 1 - P\{Y \leq 2\} \\ &= 1 - \sum_{i=0}^2 \frac{(4.5)^i}{i!} e^{-4.5} = 0.8264. \end{aligned}$$

Summary

- Discrete Random Variable, **Expectation** and **Variance**
- **Binomial** distribution: $\mathbb{E}(X) = np$, $\text{Var}(X) = np(1 - p)$.
- **Poisson** distribution: $\mathbb{E}(X) = \lambda$, $\text{Var}(X) = \lambda$.
- **Geometric** distribution: $\mathbb{E}(X) = 1/p$, $\text{Var}(X) = (1 - p)/p^2$.
- **Hypergeometric** distribution:

$$\mathbb{E}(X) = n \cdot \frac{m}{N}, \quad \text{Var}(X) = n \cdot \frac{m}{N} \left(1 - \frac{m}{N}\right) \left(1 - \frac{n-1}{N-1}\right).$$

- **Approximating** binomial distribution by using Poisson distribution

Continuous Random Variables

Recall that formally, a random variable is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

and a **discrete random variable** is a random variable that can only take a finite or at most countably infinite number of values.

We now study **continuous random variable**: X takes continuous values.

For a continuous random variable, the role of the frequency function is taken by a **probability density function** (pdf, for short), $f(x)$, such that

- $f(x) \geq 0$;
- $\int_{-\infty}^{\infty} f(x) dx = 1$.

If X has probability density function f , then the probability that $X \in A$ is

$$P\{X \in A\} = \int_A f(x) dx, \quad \text{i.e., the area under } f \text{ over region } A.$$

For $A = [a, b]$,

$$P\{a \leq X \leq b\} = \int_a^b f(x)dx = F(b) - F(a).$$

Thus

$$P\{X = a\} = \int_a^a f(x)dx = 0.$$

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Immediately, we have

$$P\{a < X < b\} = P\{a \leq X < b\} = P\{a < X \leq b\} = P\{a \leq X \leq b\}.$$

Example

Suppose that X is a continuous random variable with pdf

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2; \\ 0 & \text{otherwise.} \end{cases}$$

Find the value of C and $P\{X > 1\}$.

[Solution:]

$$1 = \int_{-\infty}^{\infty} f(x) dx = C \int_0^2 (4x - 2x^2) dx = C \left[2x^2 - \frac{2}{3}x^3 \right]_0^2 = C \left(8 - \frac{16}{3} \right)$$

and thus $C = 3/8$.

$$P\{X > 1\} = \int_1^{\infty} f(x) dx = \int_1^2 \frac{3}{8}(4x - 2x^2) dx = \frac{1}{2}.$$

For $A = (-\infty, x]$,

$$F(x) = P\{X \leq x\} = \int_{-\infty}^x f(t)dt.$$

Differentiating it, we have

$$\frac{d}{dx}F(x) = \frac{d}{dx} \left[\int_{-\infty}^x f(t)dt \right] = f(x).$$

Expectation and Variance

Definition:

Suppose that X is a continuous random variable with pdf $f(x)$. The expectation of X is defined as

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx,$$

provided that $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$.

The variance of X is defined as

$$\text{Var}(X) = \mathbb{E}(|X - \mathbb{E}(X)|^2).$$

Basic Properties:

- $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b;$
- $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2;$
- $\text{Var}(aX + b) = a^2 \text{Var}(X).$

Example

Find $\mathbb{E}(X)$ and $\text{Var}(X)$ when the density function of X is

$$f(x) = \begin{cases} 2x & x \in [0, 1]; \\ 0 & \text{otherwise.} \end{cases}$$

[Solution:]

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_0^1 x \cdot 2x dx = \frac{2}{3}.$$

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 \cdot 2x dx = \frac{1}{2}.$$

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

Theorem:

If X is a continuous random variable with pdf $f(x)$, then for any real-valued function g ,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx,$$

provided that $\int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty$.

A **uniform random variable** on the interval $[0,1]$ is a model for what we mean when we say “choose a number at random between 0 and 1.”

The density function in this case, i.e., the **uniform density** on $[0, 1]$, is

$$f(x) = \begin{cases} 1 & \text{if } x \in [0, 1]; \\ 0 & \text{otherwise.} \end{cases}$$

In **general**, the uniform density on $[a, b]$ is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b]; \\ 0 & \text{otherwise.} \end{cases}$$

and the **cumulative distribution function** is

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0 & \text{if } x \in (-\infty, a); \\ \frac{x-a}{b-a} & \text{if } x \in [a, b]; \\ 1 & \text{if } x \in [b, +\infty). \end{cases}$$

Example

If $X \sim \text{Unif}(0, 10)$, calculate the probability that

- (a) $X < 3$,
- (b) $X > 6$ and
- (c) $3 < X < 8$.

Solution:

(a)

$$P\{X < 3\} = \int_0^3 \frac{1}{10} dx = \frac{3}{10}$$

(b)

$$P\{X > 6\} = \int_6^{10} \frac{1}{10} dx = \frac{2}{5}$$

(c)

$$P\{3 < X < 8\} = \int_3^8 \frac{1}{10} dx = \frac{1}{2}$$

Theorem:

If $X \sim \text{Unif}(a, b)$, then $\mathbb{E}(X) = \frac{a+b}{2}$ and $\text{Var}(X) = \frac{(b-a)^2}{12}$.

Proof: We calculate $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$ first:

$$\mathbb{E}(X) = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2}.$$

$$\begin{aligned}\mathbb{E}(X^2) &= \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{1}{b-a} \times \frac{b^3 - a^3}{3} = \frac{b^2 + ab + a^2}{3}.\end{aligned}$$

Thus, we have

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \frac{(b-a)^2}{12}.$$

Normal Distribution

The density function of the **normal distribution** depends on two parameters μ and σ (where $-\infty < \mu < \infty, \sigma > 0$), is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

The parameters μ and σ are called the **mean** and **standard deviation** of the normal density.

- Normal Distribution is also called the **Gaussian distribution**, which plays a central role in probability and statistics.

Normal Distribution

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} dx = 1.$$

- Later in Chapter 8, we will discuss the **central limit theorem** (CLT), which justifies the applications of the normal distribution in many applications.

CLT: the sum of a large number of independent random variables is **approximately normally distributed**.

- $X \sim N(\mu, \sigma^2)$: X follows a normal distribution with parameters μ and σ .
- For $\mu = 0$ and $\sigma = 1$, we have the **standard normal density**, $N(0, 1)$. The probability density function for $N(0, 1)$ is

$$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

We use Φ to denote the cumulative distribution function for variable $N(0, 1)$:

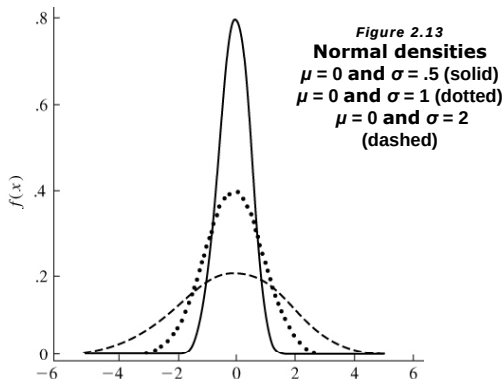
$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Normal Distribution

The density function is symmetric about μ , and so

$$f(\mu - x) = f(\mu + x).$$

The standard deviation, σ , determines the rate at which the density falls off.



Example

Suppose $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$ where $a > 0$. Find $f_Y(y)$.

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{aX + b \leq y\} \\ &= P\left\{X \leq \frac{y-b}{a}\right\} = \int_{-\infty}^{\frac{y-b}{a}} f_X(t) dt. \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} \int_{-\infty}^{\frac{y-b}{a}} f_X(t) dt = \frac{1}{a} f_X\left(\frac{y-b}{a}\right). \\ &= \frac{1}{a} \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{\frac{y-b}{a} - \mu}{\sigma}\right)^2\right] \\ &= \frac{1}{a\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{y - (b + a\mu)}{a\sigma}\right)^2\right]. \end{aligned}$$

In the example on the previous slide, we see that Y has a normal distribution with parameters $a\mu + b$ and $a\sigma$.

Theorem

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.

This proposition allows us to compute the probability $P(x_0 < X < x_1)$ where $X \sim N(\mu, \sigma^2)$ by expressing it in terms of the probability of a standard normal distribution, which [we can read off the table](#).

Example

Let $Z = \frac{X - \mu}{\sigma}$. Then by previous proposition, $Z \sim N(0, 1)$ and

$$\begin{aligned} F_X(x) &= P\{X \leq x\} = P\left\{\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right\} \\ &= P\left\{Z \leq \frac{x - \mu}{\sigma}\right\} = \Phi\left(\frac{x - \mu}{\sigma}\right). \end{aligned}$$

Thus

$$P(x_0 < X < x_1) = F_X(x_1) - F_X(x_0) = \Phi\left(\frac{x_1 - \mu}{\sigma}\right) - \Phi\left(\frac{x_0 - \mu}{\sigma}\right).$$

Example

Let $X \sim N(\mu, \sigma^2)$. Find the probability that X is less than σ away from μ , i.e., find $P(|X - \mu| < \sigma)$.

Solution:

$$P\{-\sigma < X - \mu < \sigma\} = P\left\{-1 < \frac{X - \mu}{\sigma} < 1\right\} = P\{-1 < Z < 1\},$$

where Z follows the standard normal distribution, i.e., $Z \sim N(0, 1)$.

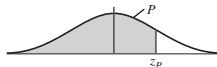
From the table, we have

$$P(-1 < Z < 1) = \Phi(1) - \Phi(-1) \approx 0.8413 - (1 - 0.8413) = 0.6826.$$

This gives $P(-\sigma < X - \mu < \sigma) \approx 0.6826$.

This shows that a normal random variable is within 1 standard deviation of its mean with probability ≈ 0.68 .

TABLE 2 Cumulative Normal Distribution—Values of P Corresponding to z_p for the Normal Curve



z is the standard normal variable. The value of P for $-z_p$ equals 1 minus the value of P for $+z_p$; for example, the P for -1.62 equals $1 - .9474 = .0526$.

z_p	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817

Example

Suppose X is a random variable with $X \sim N(20, 25)$. Find the probability that X satisfies $14 < X < 28$.

Solution: We have $\mu = 20$ and $\sigma = 5$. First, we calculate the wanted probability by using the standard normal distribution.

$$\begin{aligned} P\{14 < X < 28\} &= P\left\{\frac{14 - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{28 - \mu}{\sigma}\right\} \\ &= P\left\{\frac{14 - 20}{5} < \frac{X - 20}{5} < \frac{28 - 20}{5}\right\} \\ &= P\{-1.2 < Z < 1.6\}. \end{aligned}$$

From the table, we have

$$\begin{aligned} P\{-1.2 < Z < 1.6\} &= \Phi(1.6) - \Phi(-1.2) \\ &= 0.9452 - (1 - 0.8849) = 0.8301. \end{aligned}$$

Theorem:

Let $X \sim N(\mu, \sigma^2)$. Then $\mathbb{E}(X) = \mu$ and $\text{Var}(X) = \sigma^2$.

Proof: We first find $\mathbb{E}(Z)$ and $\text{Var}(Z)$ for $Z \sim N(0, 1)$.

$$\mathbb{E}(Z) = \int_{-\infty}^{\infty} xf(x)dx = 0$$

because the integrand is an odd function.

$$\begin{aligned}\text{Var}(Z) = \mathbb{E}(Z^2) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(-xe^{-\frac{x^2}{2}} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx \right) = 1\end{aligned}$$

Thus,

- $\mathbb{E}(X) = \mathbb{E}(\sigma Z + \mu) = \sigma \mathbb{E}(Z) + \mu = \mu$;
- $\text{Var}(X) = \text{Var}(\sigma Z + \mu) = \sigma^2 \text{Var}(Z) = \sigma^2$.